

Rising Waters: A Machine Learning Approach to Flood Prediction

Technology Stack

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	User-friendly interface

2	Backend / Server-Side	Python Flask	Lightweight web framework
3	Database / Data Storage	CSV Dataset	Stores historical flood data
4	Machine Learning	Scikit-learn, XGBoost	Model training and prediction
5	Libraries	NumPy, Pandas	Data manipulation
6	Visualization	Matplotlib, Seaborn	Data analysis
7	Cloud	IBM Cloud	Cloud deployment
8	Version Control	GitHub	Source code management
9	Model Storage	Pickle (.pkl)	Save trained model